
Indian Institute of Technology Madras Zanzibar

School of Engineering and Science

Z2004: Database Management Systems

BS-DSAI Course Schedule — Even Semester 2026

Z2004 — Database Management Systems

Document 1 of 6 | B.S. Data Science and Artificial Intelligence, Year 2

Even Semester: 2 March 2026 – 24 May 2026 (Instruction) | End-Sem: 29 Jun – 3 Jul 2026

Dr. Innocent Nyalala

Assistant Professor of Data Science and AI

Head, SAAIL Lab | IIT Madras Zanzibar

✉ innocent@iitmz.ac.in | 🕒 Office Hours: Wednesday 2:00–4:00 PM | 📍 Room 106

Teaching Assistants

Support hours: **Friday 2:00–4:00 PM (EAT)**

Sri Advaita Varakavi | ✉ zda23b010@iitmz.ac.in

Saleh Saleh | ✉ zda23b001@iitmz.ac.in

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1. Course at a Glance

 **Pedagogical Sequence: Blueprint Before Building**

This course follows a strict **theory-first** sequence. Students learn to model data (ER diagrams, normalisation, relational algebra) before writing a single SQL statement. SQL is introduced in Week 6, once the conceptual foundations are unshakeable.

Item	Detail	Item	Detail
Course Code	Z2004	Credits	10
Structure	3-1-0-0-6-10	Year / Sem	2nd Year / Even
Lectures	3 per week, 50 min	Tutorials	1 per week, 50 min
Primary DB	SQL Server 2022	Secondary DB	PostgreSQL 16
Quiz 1	Week 6 (Apr)	Quiz 2	Week 12 (May–Jun)
End-Sem	29 Jun – 3 Jul	Project Final	22 Jun 2026

2. Assessment Structure

Component	Weight	Date	Coverage
Quiz 1	15%	Apr 16–18	M1 (ER), M2 (Relational Model), M3 (Rel. Algebra)
Quiz 2	15%	May 23–26	M4–M6 (SQL Basics to Advanced)
End-Sem Exam	40%	Jun 29 – Jul 3	All modules; emphasis M7–M10
Semester Project	20%	Milestones	RAG Pipeline or Recommendation Engine
Assignments (ICA)	10%	Fortnightly	SQL puzzles, schema sketches, think-pair-share
Total	100%		

3. Module Overview

#	Module	Weeks	Key Topics
M1	Foundations and Data Modelling	1–2	DBMS intro, ER model, weak entities, extended ER
M2	Relational Model	3	Relations, schemas, keys, integrity constraints
M3	Relational Algebra	4–5	σ , π , $\bowtie$, set ops, division
M4	SQL Foundations	6–7	DDL, DML, SELECT, WHERE, ORDER BY
M5	SQL Intermediate	8–9	Aggregation, GROUP BY, joins, subqueries
M6	SQL Advanced	10	CTEs, window functions, triggers, stored procs
M7	Normalisation and FD Theory	11–12	FDs, Armstrong’s axioms, 1NF–BCNF, 4NF
M8	Storage and Indexing	13	B-Tree, hashing, heap files, buffer pool
M9	Query Processing and Transactions	14	Query pipeline, ACID, 2PL, recovery
M10	Modern and AI Data Systems	15–17	NoSQL, vector DBs, embeddings, RAG pipelines

4. Week-by-Week Schedule

 **Holiday Notice**

Eid al-Fitr: 19–20 Mar (Thu–Fri). **Good Friday:** 3 Apr. **Easter Monday & Karume Day:** 6–7 Apr. **Union Day:** 26 Apr (Sun — no impact). **Labour Day:** 1 May (Fri). **Eid al-Adha:** 27 May (Wed).

Sessions adjacent to holidays are rescheduled as noted below.

Week	Dates	Topic	Activity / Notes
		Sess.	
		L1 Course Introduction; File vs. DBMS	Why flat files fail; three-schema architecture

Week	Dates	Sess.	Topic	Activity / Notes
1	2–6 Mar	L2	DBMS Architecture and Roles	Components; DBA vs. end-user; data independence
		L3	Data Models Survey	Hierarchical, network, relational, object — visual timeline
		T1	Lab 1: Environment Setup	Install SQL Server 2022 + SSMS; explore sample DB
		L4	ER Model: Entities and Attributes	Strong/weak entities; simple/composite/multivalued attr
2	9–13 Mar	L5	ER Model: Relationships	Cardinality ratios; participation; role names
		L6	Extended ER: ISA, Aggregation	Specialisation; aggregation design decisions
		T2	Lab 2: ER Design Practice	Design ER for university; draw in draw.io; map to tables
		L7	Relational Model Fundamentals	Relation; tuple; domain; schema; instance
		L8	Integrity Constraints	Entity, referential, domain; superkey; candidate key
		L9	ER to Relational Mapping	All 8 mapping rules; worked hospital example
3	16–20 Mar	T3	Lab 3: Mapping and DDL	Map ER to DDL; form groups; short-list project topics
		L10	Relational Algebra I	σ , π , ρ , union, intersection, difference
		L11	Relational Algebra II	Natural join, theta join, outer joins; expression trees
		L12	Relational Algebra III	Division; query equivalence; ICA: write algebra for 5 queries
4	23–27 Mar	T4	Lab 4: Algebra Drill	Translate 15 English queries to algebra; verify with SQL; Milestone 0: Registration due (27 Mar)
		L13	Tuple Relational Calculus	TRC syntax; safe expressions; equivalence with algebra
		L14	Domain Relational Calculus	DRC; QBE visual language; comparison with TRC
		L15	SQL Introduction: DDL	CREATE, ALTER, DROP; data types in T-SQL
5	30 Mar–3 Apr	T5	Lab 5: First SQL Statements	Write DDL for university DB; insert 20 rows; query structure

Week	Dates	Sess.	Topic	Activity / Notes
6	6–10 Apr	L16	SQL DML: Basic SELECT	SELECT, FROM, WHERE, ORDER BY, DISTINCT
		L17	SQL DML: INSERT, UPDATE, DELETE	INSERT, UPDATE, DELETE; TRUNCATE; transactions
		QUIZ 1 — Thu 16 Apr & Fri 17 Apr		Coverage: M1–M4. 60 min. Closed book.
		T6	Lab 6: DML Practice	20-query lab on university DB; INSERT/UPDATE/DELETE tasks; Milestone 1: Schema & DDL due (10 Apr)
7	13–17 Apr	L18	SQL Aggregation	COUNT, SUM, AVG, MAX, MIN; NULL handling
		L19	GROUP BY and HAVING	GROUP BY; HAVING; execution order; common errors
		L20	SQL JOINS (Inner, Outer, Self)	INNER JOIN, LEFT/RIGHT/FULL OUTER JOIN, SELF JOIN; visual diagrams
		T7	Lab 7: Aggregation and Joins	3 debugging exercises; sales summary report; Quiz 1 results returned
		L21	Subqueries	Scalar, row, table; IN, EXISTS, ALL, ANY
		L22	Correlated Subqueries	Step-by-step execution trace; vs. join performance
		L23	Views and Indexes	CREATE VIEW; updatable views; CREATE INDEX
		T8	Lab 8: Subqueries and Views	Rewrite 5 queries using subqueries; create and test views
		L24	CTEs and Recursive Queries	WITH; recursive CTE; org-chart / hierarchy example
10	11–15 May	L25	Window Functions I	ROW_NUMBER, RANK, DENSE_RANK, NTILE
		L26	Window Functions II	LAG, LEAD, SUM OVER, PARTITION BY
		T9	Lab 9: Advanced SQL Challenge	Competitive 5-problem SQL contest; window + CTE tasks
		L27	Stored Procedures and Triggers	CREATE PROCEDURE; parameters; CREATE TRIGGER; AFTER/INSTEAD OF

Week	Dates	Sess.	Topic	Activity / Notes
		L28	Transactions in SQL	BEGIN TRAN, COMMIT, ROLLBACK; isolation levels
		L29	SQL Security	GRANT, REVOKE; roles; row-level security; SQL injection
		T10	Lab 10: Procs, Triggers, Transactions	Write 3 stored procedures; simulate deadlock; Milestone 2: Dataset & queries due (15 May)
		L30	Functional Dependencies	Definition; trivial/non-trivial; closure algorithm
		L31	Armstrong's Axioms and Canonical Cover	Reflexivity, augmentation, transitivity; canonical cover steps
		L32	Normal Forms 1NF – BCNF	Anomalies illustrated; 1NF, 2NF, 3NF, BCNF with examples
		T11	Lab 11: Normalisation Drill	Decompose 5 unnormalised schemas to BCNF
		L33	4NF, 5NF and Decomposition	Multivalued deps; lossless-join test; dependency preservation
		L34	Storage: Disk, Files, and Buffer Pool	Storage hierarchy; heap files; slotted pages; buffer manager
		QUIZ 2 — Sat 23 May & Tue 26 May		Coverage: M4–M7. 75 min. One A4 cheat sheet.
13	1–5 Jun	T12	Lab 12: Design Theory Synthesis	Infer FDs from CSV dataset; normalise; generate DDL
		L35	B-Tree and B ⁺ -Tree Indexing	Structure; search/insert/delete; clustered vs. unclustered
		L36	Hashing and Secondary Indexes	Static/dynamic hashing; dense/sparse indexes; index selection
		L37	Query Processing Pipeline	Parsing; relational algebra translation; cost estimation
		T13	Lab 13: Index Performance	Measure query time before/after indexing; read EXPLAIN ANALYZE; Milestone 3: Performance evidence due (5 Jun)
14	8–12 Jun	L38	Join Algorithms and Query Optimisation	NL-join, sort-merge, hash join; heuristic vs. cost-based
		L39	Transaction Model and ACID	Transaction states; ACID; schedules; serial vs. concurrent

Week	Dates	Sess.	Topic	Activity / Notes
		L40	Concurrency Control	Conflict serializability; 2PL; deadlock detection
		T14	Lab 14: Concurrency Simulation	Trigger deadlock; observe lock monitor; resolve
		L41	Crash Recovery	Log-based recovery; undo/redo; checkpoints; ARIES overview
		L42	NoSQL Paradigms	CAP theorem; key-value, document, column-family, graph
		L43	Vector Databases and Embeddings	Cosine similarity; HNSW; pgvector; Pinecone
		T15	Lab 15: MongoDB and Vector Search	Insert documents; run aggregation; embed 500 docs; nearest-neighbour query
		L44	RAG Pipeline Architecture	Ingest-chunk-embed-store-retrieve-generate; full diagram
		L45	Cloud Databases (One Lecture)	AWS RDS, Azure SQL, Google Cloud SQL, Supabase; cost models
	Thu 24 Jun: Last Day of Instruction.			Course review and Q&A. Final submission
		T16	Lab 16: RAG Pipeline	Build mini RAG using Python + pgvector; test question-answering

5. Project Milestone Summary

MS	Due	Deliverable	Weight	Notes
M0	Fri 27 Mar 2026	Group Registration + Topic	0%	Mandatory; late registrations lose 10 marks
M1	Fri 10 Apr 2026	Schema + DDL Script	20%	ER diagram + <code>schema.sql</code> runs cleanly
M2	Fri 15 May 2026	Dataset + Queries	20%	<code>data.csv/json</code> + <code>queries.sql</code> + <code>README.md</code>
M3	Fri 5 Jun 2026	Performance Evidence	20%	Max 5-page PDF + <code>performance.sql</code>
Final	Mon 22 Jun 2026	Full Submission	40%	GitHub link + 5-minute demo video + Q&A

6. End-Semester Examination Schedule

Session	Date	Coverage
End-Semester Exam	Week of 29 Jun – 3 Jul	Single 3-hour sitting, 100 marks. All 10 modules. Section A (20 marks): MCQ; Section B (25 marks): ER, Normalisation, RA; Section C (30 marks): SQL Problem Solving (HospitalDB schema); Section D (25 marks): Transactions, NoSQL, Vector DBs, RAG.

💡 Examination Note

The End-Semester Examination is a **single, continuous 3-hour sitting**. All four sections (A, B, C, D) are answered in the same session on the same day. The exact date within the window 29 Jun – 3 Jul 2026 will be communicated no later than two weeks before the exam.

Subject to revision. Changes announced via LMS with 48 hrs notice.
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